

Subject Description Form

Subject Code	ME6102
Subject Title	Advanced Topics in Control, Acoustics, and Dynamics
Credit Value	3
Level	6
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	1. To equip students with the knowledge of advanced control systems. 2. To equip students with the knowledge of advanced acoustics. 3. To equip students with the knowledge of advanced wave theory.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a) Model the dynamic behaviour of mechanical systems and analyse their properties. b) Synthesise feedback control methods for automating various systems and machines. c) Model acoustic wave generation, propagation, reflection, scattering and absorption in various media. d) Apply the conceptual and theoretical acoustics knowledge to analyse acoustical problems. e) Apply the wave propagation theory to simulation, analyse and solve problems of wave propagation. f) Design non-destructive evaluation approaches using guided waves.
Subject Synopsis/ Indicative Syllabus	Module 1. Automatic Control Systems. • Systems Modelling. Discrete/continuous-time systems; Linear/non-linear systems; Energy-based dynamics (Lagrangian, Hamiltonian); Differential kinematic systems. • Automatic Control Systems. Feedback signals; Lyapunov stability; Optimal control; Passivity-based control; Model-based nonlinear control; Adaptive control systems; Sensor-based control; Servomechanisms; Multi-agent systems; Learning-based control. Module 2. Advanced Acoustics • Acoustics. Dynamics of fluid motion; Acoustic wave motion; Sound speed; Green's function; Acoustic modes; Sound transmission and absorption; Sound attenuation. • Structural acoustics. Vibration of continuous system; coupling between sound and vibration; Sound radiation; Sound transmission through a finite plate; Periodically supported systems. • Numerical methods in acoustics. Modal analysis; Finite element method; Computer simulations. Module 3. Elastic Wave Propagation in Solids: Linear and Nonlinear Aspects

	• Fundamental Theory. Dispersion; Waves in layered plates and hollow cylinders; waves in viscoelastic medium. • Modelling Waves. Analytical solutions; semi-analytical finite element method; modeling wave propagation. • Nonlinear Method. Bulk waves in weakly nonlinear elastic medium; second harmonic generation; Weakly nonlinear wave equation. • Study Extension. Phased array; imaging.																																						
Teaching/Learning Methodology	Lectures and Tutorials																																						
Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="6">Intended subject learning outcomes to be assessed (Please tick as appropriate)</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th><th>e</th><th>f</th></tr><tr><td>1. Continuous Assessment</td><td>50%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>2. Examination</td><td>50%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>Total</td><td>100 %</td><td colspan="6"></td></tr></table> Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: 1. The assessment is comprised of 50% continuous assessment and 50% examination. 2. The continuous assessment aims at evaluating the progress of students study, assisting them in self-monitoring of fulfilling the respective subject learning outcomes, and enhancing the integration of the knowledge learnt. 3. The examination is used to assess the knowledge acquired by the students for understanding and analyzing the problems critically and independently; as well as to determine the degree of achieving the subject learning outcomes.	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)						a	b	c	d	e	f	1. Continuous Assessment	50%	✓	✓	✓	✓	✓	✓	2. Examination	50%	✓	✓	✓	✓	✓	✓	Total	100 %						
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Reading List and References	1. Lecture Notes 2. Allan Pierce, Acoustics: an introduction to its physical principles and applications, Latest Edition 3. Jean-Jaques Slotine, Applied Nonlinear Control, First Edition 4. Philip Morse, Uno Ingard, Theoretical Acoustics, Latest Edition 5. Fabien Anselmet, Pierre-Olivier Mattei, Acoustics, Aeroacoustics and Vibrations, Latest Edition 6. Joseph Rose, Ultrasonic Guided Waves in Solid Media, Cambridge University Press, Latest Edition
Last Update	June 2024